
LAB ASSESSMENT 1 - GROUP OF TWO
Features Tracking for Web-based Augmented Reality

*Date Issued: 17 Mar 2026**ajune@utm.my***OVERVIEW**

This lab activity is a part of the course assessment series that aims to determine a standard feature-based tracking technique that enables the creation of augmented reality (AR) web-based applications. It uses computer vision technology to recognize and track the images in real-time and overlays the 3D object (as simple as 3D cube). This project brings AR feature-based marker tracking technique to the Web. Feature-based tracking is commonly used to build AR application while Web is the most ubiquitous computing platform. There are already billions of devices connected to the web that have access to a plethora of visual information. OpenCV is a library for general computer vision application development, with hundreds of algorithms and efficient implementation. With HTML5 and new client-side technologies today browsers are capable of handling more complex tasks. You are required to develop a web application to produce AR marker using a feature-based tracking technique. A captured image is converted into a grayscale image, and the grayscale image has created the features.



Then your application is able to position and orient virtual objects, such as 3D cube on that generated features. Once your web-camera captures the image, the features are recognized and 3D object will appear on the top of it.

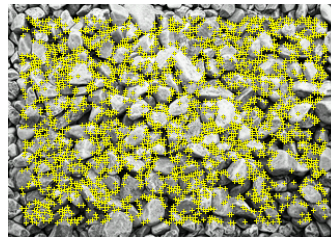
INSTRUCTIONS TO STUDENTS:

1. This project **MUST** be done in a group (a group of two) and completed before the deadline.
2. You need to demonstrate a RGB image is converting to features using your web browser application. The testing involves at least 3 images, presenting the results of 3 images is different 3D object in your video output.
3. You need to run ONE (1) AR scene using your generated marker. You need a 3D object to overlay on the top of your marker output.
4. The following requirements are the bare minimum for the assignment:

- Hardware
 - **Printed marker** – A4 size, a coloured image. (**do not** use mobile screen or window screen as your marker, **do not** use black and white as a marker)
 - **Web-camera** – Display device (your web application should work on desktop-based)
- Tracking method – Natural feature tracking



[Image]



[Features]

- Events/Actions
 - **Upload Image** button – to upload RGB image
 - **Generate** button – to execute the feature extraction method (converting image to grayscale and generate features)
 - **Save** button – to save the current image (in acceptable format)
5. You are required to present your project DEMO (video format) and prepare ONE (1) page description about your AR web-based application and your feature-based marker generator.
 6. To guide you more :
 - a. Computer vision using Javascript : TrackingJS (<https://trackingjs.com/>)
 - b. JSFeat (<http://inspirit.github.io/jsfeat/>)
 - c. OpencvJS (<https://github.com/ucisysarch/opencvjs>)
 - d. **AR.js** - based on ARToolkit marker : <https://github.com/jeromeetienne/AR.js>
 - e. **ARToolkit.js**: <https://github.com/kalwalt/jsartoolkit5>
 - f. Feature extraction method - [SURE, ORB, SIFT, BRIEF](#).
 - g. Some various experiments with **Ar.js** and **Three.js** : <https://kalwalt.github.io/kalwalt-interactivity-AR/index.html>

SUBMISSION PROCEDURE:

The **DEADLINE** given as in **EDREDO** submission link, group leader needs to perform submission .ZIP contains AR website, video and one page report to email ajunewanis@ymail.com

In any reasons the group failed to deliver the DEMO and any of each member in a group failed to demonstrate verbally and/or technically on their deliverable, the mark WILL BE ENDED WITH A ZERO MARK FOR BOTH PARTY.

Assessment Criteria	Marks
You need to demonstrate a RGB image is converting to features using your web application Not creating your tracking method – if the students use a ready-made library [Deduction 50 Marks]	10
The testing involves at least 3 images, presenting the results of 3 images in your video output. Fail to adhere, present only one (1) image [Deduction 3 Marks]	6
You need to run ONE (1) AR scene using your generated marker. Fail to adhere [Deduction 3 Marks] Not using your own generated marker [Deduction 3 Marks]	3
You need a 3D object to overlay on the top of your marker output. Overlay a video, 2D elements [Deduction 3 Marks]	3
Printed marker – A4 size, a coloured image. Not A4 size [Deduction 3 Marks] Not a coloured image [Deduction 3 Marks]	3
Do not use mobile screen or window screen as your marker Fail to adhere [Deduction 5 Marks]	1
Do not use black and white as a marker Fail to adhere [Deduction 5 Marks]	1
Use web-camera and not deliver a desktop-based AR application Fail to adhere [Deduction 5 Marks]	1
UI – Web application (Actions and events) Run without UI [Deduction 10 Marks]	3
Video formatting – DEMO Inappropriate demo-reel, corrupted or void, missing link [Deduction 2 Marks]	2
ONE (1) page report – Report written in more than 1 page [Deduction 2 Marks] Report has been sent without name and matrix ID [Deduction 5 Marks] Deliver without a report [Deduction 15 Marks]	5
Upon completion this assignment the students will attain a team-working marks	10
Submission ethic bonus: Submission more than ONE (1) time to ymail [Deduction 2 Marks] Missing the deadline [Deduction 50 Marks]	2
Total	50

* Deduction marks over the total mark. If the total mark produces a negative value, his assignment is automatically ended with ZERO marks.